

When Order and Time Disagree: Position Overrides Explicit Time in Language Models

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Abstract

When a context states several values of the same quantity—a variable rewritten in a chain of thought, a user’s city across chat sessions, a setting across commits—a language model must decide which value is current. We show that when the order of presentation and the explicit time disagree, open models largely follow the order. In shuffled chains of thought that overwrite a variable eight times, all 14 models we tested at 1.7B–14B parameters answer with the last-presented value in 89.5–99.5% of items, and Qwen3-32B does so in 97%. On MemConflict, a third-party benchmark of conflicting user states, showing the same dated sessions newest first costs Qwen3 models 21–40 accuracy points from 4B to 32B, and merely ordering them by retrieval relevance costs 10–19 points. Scale helps only where dates are close to the facts: on newest-first logs, agent memories and short conversations, Qwen3 models of 14B and above lose at most 7 points, whereas sessions whose dates sit in headers thousands of tokens before the facts still cost 21–24 points at 32B. A second model family, Gemma 3 12B, shows it on MemConflict more strongly than Qwen3 of similar size (preliminary). The effect is specific: on PersonaMem and LoCoMo temporal questions, which do not require choosing between an older and a newer state, order has no effect at any size. We trace two failure points: in short contexts the model represents which value is newer but does not use it, and copies the misleading order into its own reasoning; in long sessions, dates are not bound to the facts they govern. Prompts and ordering statements do not help. Training a 4B model, on synthetic block-dated documents in genres it is never tested on, to list the dated records of the queried attribute inside its thinking removes the order effect on held-out ConvoMem-long (newest first 77→97) and on MemConflict (36→78), while PersonaMem stays within 3 points of the base model.

1 Introduction

Much of what language models read is a record of change. A chain of thought (CoT) assigns a value and later revises it; a chat assistant’s memory holds the user’s address before and after a move; a log lists the successive values of a configuration key. To answer “what is the value now?”, a reader must order these statements in time. In most training text this is trivial, because text is written in the order events happen: the statement written last is also the most recent. *Later in the text* is therefore an excellent proxy for *later in time*.

Many contexts that models read at deployment break this proxy. E-mail clients, `git log`, changelogs and feeds list the newest entry first. Retrieval-augmented memories return passages by relevance, not by date. Agents append retrieved memories after their own trajectory; monitors read other models’ reasoning out of order; tools concatenate outputs from many sources. In all of these,

Chronological	Newest first	Retrieval order
oldest first	inbox, git log, feeds	most relevant first (BM25)
Session (2024-03-02) User: I live in Darwin.	Session (2024-09-14) User: I just moved to Melbourne.	Session (2024-09-14) User: I just moved to Melbourne.
Session (2024-09-14) User: I just moved to Melbourne.	Session (2024-03-02) User: I live in Darwin.	Session (2024-03-02) User: I live in Darwin.
Q: Where does the user live now?	Q: Where does the user live now?	Q: Where does the user live now?
Last-written value: Melbourne correct	Last-written value: Darwin stale (dates ignored)	Last-written value: Darwin stale (dates ignored)

Figure 1: Order versus time. The same two dated sessions shown in three orders (schematic example). A reader that takes the last-written value as current answers correctly only under chronological presentation. Newest-first lists and retrieval-ranked memories are common at deployment; the dates are present in all three cases.

time is usually still *written down*—as a date, a timestamp or a step number—but it disagrees with the order of presentation (Figure 1).

This paper asks a simple question: **when presentation order and explicit time disagree, which one does a language model follow?** Standard benchmarks cannot answer it, because they present updates chronologically, so that a reader who uses dates and a reader who uses position give the same answer (§3). We therefore hold the content fixed and vary only the order, across controlled chains of thought, deployed formats, and third-party long-term memory benchmarks, and across model sizes from 1.7B to 32B parameters.

Our findings are:

1. **Position wins over explicit time.** In shuffled chains of thought, readers answer with the last-presented value in 89.5–99.5% of items once a variable is overwritten eight times; a step number on every line overrides this only in the larger Qwen3 models, a “listed from most recent” header rarely does, and an explicit retraction (“Wait, that’s wrong.”) only partly offsets it (§4.1). The same default appears in deployed formats: showing dated LongMemEval sessions newest first costs 24–53 points, and stating that the sessions are ordered newest first recovers at most 2.6 (§4.2).
2. **It reproduces on third-party benchmarks, and natural retrieval order triggers it.** On MemConflict [42], Qwen3-4B drops from 75.4% to 36.2% when sessions are shown newest first, and to 56.2% when they are shown in BM25 relevance order—an order that a standard retrieval memory produces without any adversarial intervention (§4.3).
3. **Scale removes the effect only where dates are local.** From Qwen3-4B to Qwen3-32B the newest-first cost shrinks from 11 points to 3 on dated logs and from 18 to 7 on 2.4k-token conversations, but it remains 23–24 points on LongMemEval at every size and 21 points on MemConflict at 32B (40 at 4B), where dates sit in session headers far from the facts. Larger models use step numbers in short chains of thought (32B: 80% correct, 4B: 4%), yet still read the last-presented value when no cue is given (97%). Gemma 3 12B shows the effect on MemConflict and on newest-first changelogs more strongly than Qwen3 of similar size (§5).
4. **It is specific to conflicting states.** On PersonaMem [20] and LoCoMo [31] temporal questions, order has no effect at any size. Diagnostics show that these questions do not require choosing

between an older and a newer state (PersonaMem’s update questions can even be answered without the history), consistent with our CoT finding that order matters only when a state is overwritten (§6).

5. **Two failure points.** In short contexts the failure is one of *selection*: the newer value is linearly decodable at the answer position in $\approx 98\%$ of items, including those the model answers with the stale value, and the misleading order is copied into the model’s own timeline or thinking, which it then reads by position. In long sessions the failure is one of *binding*: dates in session headers are not attached to facts that appear thousands of tokens later. Questions about the *earliest* value show no sign of a semantic “written later means newer” rule in any of six models; we therefore claim only that position overrides explicit time (§7).
6. **A fix that never sees the test formats.** Pre-registered decoupled training shows that the selection failure is one of use, not knowledge, but answer-only training damages the answer format and, at 1.7B, arithmetic reasoning. Training the model to write a dated list of the relevant records inside its thinking, on synthetic documents with no chat, e-mail or log formats, removes the order effect on held-out ConvoMem-long and on MemConflict while keeping PersonaMem within 3 points of the base model (§8).

For the training study and for every experiment after the initial chain-of-thought study, we wrote a numerical threshold before running it; we report all outcomes, including the many that did not meet their thresholds (Appendix A), and every version of the fix that we evaluated on held-out data (Appendix D).

2 Related work

Order and position in context. Models use information at the beginning and end of long contexts better than in the middle [29], and permuting premises hurts reasoning [7]. Studies of shuffled chains of thought report that answers largely survive sentence- or line-level shuffling [8, 44]. Closest to our chain-of-thought results, Liu [28] show that small (1–3B) models read arithmetic CoT by copying whichever number occupies the trailing position, which is why shuffled arithmetic CoT keeps most of its accuracy, while shuffling breaks BBH tracking and logical deduction and larger models become more selective. Our CoT experiments extend this positional readout to states that are overwritten several times and pit it against explicit time cues; unlike them, we find logical deduction almost unaffected by shuffling (§4.1), consistent with our account that order matters when a state is overwritten. For many in-context updates of a key–value list, retrieval of the latest value degrades and early values intrude [6, 37]; with up to 64 overwrites inside a chain of thought we instead observe a stable recency default. Fitch [14] find that source framing outweighs position for two short conflicting documents, with a primacy rather than a recency bias for Gemma; Fatemi et al. [13] show that fact order affects temporal reasoning; and in-context retrieval shows recency and contiguity biases reminiscent of human episodic memory [4]. When recency and another cue are perfectly correlated in training, which rule a model adopts is not determined by the data [27]; this is why we make the cues conflict. None of these studies pits presentation order against explicit time.

Knowledge updates and stale state. Entity tracking has been studied with box-moving tasks [22]; Tang et al. [41] find that models do not update world states step by step but aggregate the relevant statements at query time, and fail when the order of operations matters. Concurrently, Guo et al. [16] characterize “stale binding” when a variable is updated in context: the current value remains decodable, attention drifts to older values, and an attention bias at known positions repairs the answer. Their updates are always presented chronologically and the fix requires knowing

where the current value is; we study the conflict between order and explicit time, show the failure is copied into the model’s own reasoning, separate selection from binding in long sessions, and repair the long-session failure without position information. Outdated retrieved documents can override correct parametric answers, and dates alone barely help [1]; temporal conflicts across document versions mislead models [34]; and masking superseded records in the KV cache shifts preference toward new values at a cost in detail [50]. On the generation side, reflection markers such as “wait” behave as surface cues [48]; we find a similar weakness of retraction markers on the reading side.

Long-term memory. LongMemEval [45] includes knowledge-update questions and sorts retrieved sessions by timestamp by default; we quantify the cost of not doing so. MemConflict [42] evaluates memory systems under dynamic, static and conditional conflicts but does not vary presentation order. RoutePrism [46] varies the order in which records are *ingested* into a memory and shows that retention policies then keep different evidence; our effect arises with all evidence present. Jiang et al. [21] show that much of the gain of structured memories comes from a readable, time-ordered rendering, in line with our results. StateMemBench [12] isolates “state drift”, acting on a superseded value that is present in context, and lists “trust the most recent mention” among the lazy reader policies it is designed to trap. Memory systems add explicit state labels [38] or temporal indexes [30], and Supersede [36] trains a bounded-memory agent to keep current values, noting that frontier models resolve explicit updates by a last-mention scan—the heuristic we show fails out of chronological order. Some fixes build the same assumption in: DZ-TDPO [26] adds an attention bias that decays with token distance, assuming that valid updates sit at the trailing edge of the context, which our results suggest would hurt under newest-first or retrieval order. Long-context training on synthetic, information-dense data improves the use of the middle of the context without hurting general ability when general instruction data are mixed in [2]; our fix follows that recipe for its general data.

3 Setup

Formalization. A context contains records $r_1, \dots, r_n$ about one queried quantity. Record r_i carries a value v_i and a time τ_i that is *written in the text* (a date, a timestamp, or a step number), and the records are presented in an order given by a permutation π , where $\pi(i)$ is the position of r_i . The current value is

$$v^* = v_{i^*}, \quad i^* = \arg \max_i \tau_i. \quad (1)$$

Two idealized readers answer differently:

$$f_{\text{time}}(r, \pi) = v_{\arg \max_i \tau_i}, \quad f_{\text{pos}}(r, \pi) = v_{\arg \max_i \pi(i)}. \quad (2)$$

They agree whenever π is increasing in τ , which is the case in almost all naturally written text and in standard benchmarks. We therefore compare the same records under several orders: CHRON (oldest first, π increasing in τ), NEWEST (newest first, π decreasing in τ), shuffled, and RETR, in which records are sorted by a BM25 relevance score $s(q, r_i)$ to the question q , most relevant first:

$$\pi_{\text{retr}}(i) < \pi_{\text{retr}}(j) \iff s(q, r_i) > s(q, r_j). \quad (3)$$

RETR is not chosen by us: it is the order a standard retrieval memory hands to the reader.

Measures. We report accuracy and the *stale rate* (the answer is a superseded value, or states the change in the wrong direction). The *order effect* is the paired difference $\Delta = \text{Acc}(\text{ORDER}) - \text{Acc}(\text{CHRON})$ over the same items, with a 95% paired bootstrap interval (10,000 resamples of items). Because items in the held-out sets share contexts (ConvoMem personas, PersonaMem’s 33 shared contexts, MemConflict users, LoCoMo’s 10 conversations), we also resampled these clusters; no conclusion changes, and the intervals widen by at most 1.5 points. Free-text answers are graded by Qwen3-14B (4-bit) given the question, the reference answer and, where available, the older and newer evidence, choosing among *up to date*, *stale*, and *other*. On responses that name both values, the judge agrees with a string rule (which value is mentioned first) on 90.8% of Qwen3-4B’s and 91.2% of Qwen3-32B’s MemConflict answers; on LongMemEval, agreement with string matching is 91.4%. The same judge is used for every model.

Models and inference. Chain-of-thought experiments use 14 open models: Qwen3-1.7B/4B/4B-Base/8B/14B [47], OLMo-2-7B (Base, SFT, DPO, Instruct) and OLMo-2-13B (Base, Instruct) [33], Qwen2.5-Math-7B, DeepSeek-R1-Distill-Qwen-7B [11], and Phi-4-mini-instruct [32]; models of 7B parameters and above run in 4-bit with Hugging Face Transformers (HF). The application and mechanism experiments use Qwen3-4B, Qwen3-14B (4-bit) and Phi-4-mini on HF. The scale study (§5) runs Qwen3-4B, 8B, 14B and 32B and Gemma 3 12B [15] in bf16 on vLLM [24], with identical prompts, decoding rules and judge. vLLM reproduces HF item by item on 97.2% of short items and 94.0% of 37k-token items, with condition means within 2 points; the second figure missed our pre-registered 95% criterion, so we compare models only within one engine and mark every cross-engine comparison. All models answer in non-thinking mode with greedy decoding unless stated; thinking mode uses Qwen3’s recommended sampling (temperature 0.6, top- p 0.95, top- k 20) with a fixed seed per item.

Protocol. For the training study (pre-registered) and for all experiments after the initial chain-of-thought study, we wrote the hypothesis and a numerical threshold before running; in the chain-of-thought study, task categories and conditions were fixed in code before running. Controls differ from treatments in exactly one variable, except where we say otherwise. A phenomenon *replicates* on a set when the order effect is at most -10 points and its interval excludes zero. LongMemEval was used to discover the phenomenon and is reported as *seen*; ConvoMem personas 50–99, ConvoMem-long, PersonaMem, MemConflict and LoCoMo are held out and are never used to tune anything. We evaluated every version of our long-session fix on them, and two of them informed the *design* of later versions through aggregate failure types—MemConflict from the third version, PersonaMem from the fourth; we flag them wherever we report the fix (Appendix D). Table 1 lists the test sets.

4 Position is read as time

4.1 Chains of thought

Shuffling hurts only overwritten state. We generate 16-line programs whose CoT assigns the queried variable k times, and ask the reader for its final value after the CoT lines are shuffled (Figure 2a; Qwen3-4B, $n = 200$ per cell). Variables assigned once are read correctly after shuffling ($\geq 99.5\%$), but accuracy on overwritten variables falls to 79.5/39.5/19.0/14.0% at $k = 1/2/4/8$, close to chance among the $k+1$ candidates. The reader is not guessing: it picks the value presented last in 93.5–98.5% of items for $k \geq 2$, and in 93–99% of items in a longer variant with up to $k = 64$

Table 1: Test sets. “Conflict”: the question requires choosing between an older and a newer state of the same quantity. Orders: C = chronological, N = newest first, S = shuffled, R = BM25 retrieval order. *Held out, but aggregate failure types informed later versions of the fix (Appendix D).

Set	Source	Length (tokens)	Conflict	Orders	Status
Variable tracking in CoT	ours (synthetic)	16–72 lines	yes	C, S, N	–
BBH, model’s own CoT (26 tasks)	Suzgun et al. [39]	short	varies	C, S	–
LongMemEval knowledge update (78)	Wu et al. [45]	4.1k–9.3k	yes	C, N	seen
E-mail threads, git log, CHANGELOG	ours (synthetic)	0.5k–8.4k	yes	C, N	–
Agent trajectories + memory	ours (synthetic)	short	yes	placement	–
ConvoMem-long (124)	Pakhomov et al. [35]	2.4k (median)	yes	C, N	held out
MemConflict, dynamic (240)	Tao et al. [42]	6.7k–21.7k	yes	C, N, R	held out*
PersonaMem 32k (589)	Jiang et al. [20]	16k–30k	no	C, N	held out*
LoCoMo temporal (321)	Maharana et al. [31]	13k–25k	no	C, N, R	held out
MemoryAgentBench CR (76)	Hu et al. [19]	6.9k	counterfactual	C, N, R	seen

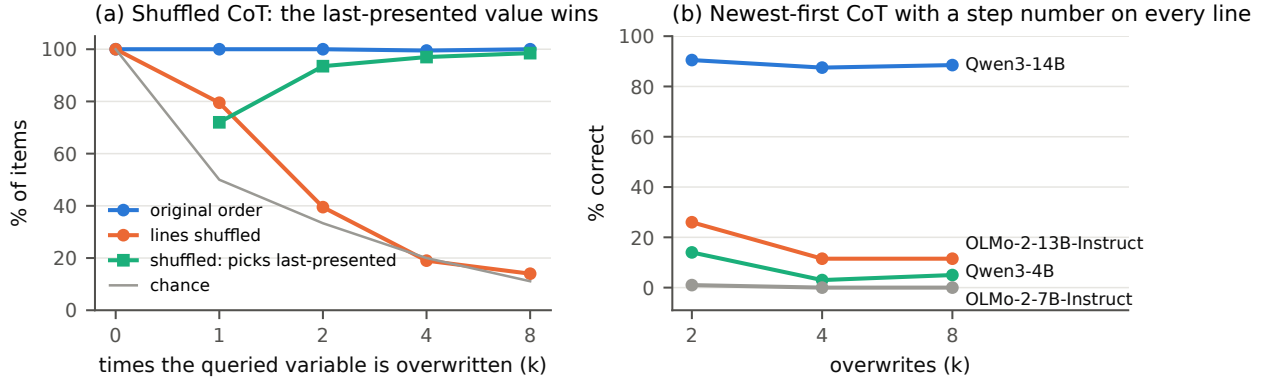


Figure 2: Chain-of-thought readers take the last-presented value. (a) Qwen3-4B on a synthetic 16-line program whose CoT overwrites the queried variable k times ($n = 200$ per point). After shuffling, accuracy falls to chance because the reader picks whichever value is presented last. (b) Newest-first CoT with an explicit step number on every line; accuracy in the conflict case, where position points to a stale value. Among these models only Qwen3-14B uses the step numbers; Figure 4b adds Qwen3-8B and 32B.

overwrites, where the first-presented value is never chosen for $k \geq 2$. At $k = 8$, all 14 models pick the last-presented value in 89.5–99.5% of shuffled items, while variables assigned once stay above 96.8%. Larger models resist at small k (at $k = 2$: Qwen3-14B 73.5%, OLMo-2-13B-Instruct 87.0%) but not at $k = 8$ (97.0% and 95.0%); Qwen3-32B also picks the last-presented value in 97% of items at $k = 8$ (§5). Base, SFT, DPO and RL-tuned OLMo-2-7B checkpoints behave alike (93.5–99.0% at $k = 8$), and a natural-language version (“Box E has 41 marbles”) behaves identically. On the model’s own reasoning for all usable BBH tasks, shuffling breaks 39% of answers on tasks that we labeled in advance as tracking updated state, and 4% on the others. This is consistent with reports that shuffled CoT preserves answers [8, 44], which concern reasoning whose states are mostly not overwritten: order matters when a state is overwritten. On logical deduction we see a small drop (five objects: 89.6%→85.3%), whereas Liu [28] report a large one; the settings differ (line-level shuffling of the model’s own CoT here, 1–3B readers there).

Explicit order cues lose to position. When the CoT lines are listed newest first with a step number on every line (Figure 2b), Qwen3-4B answers correctly in 3–14% of conflict cases, OLMo-2-13B-Instruct in 11.5–26% and OLMo-2-7B-Instruct in 0–1%; of the four models we tested

in this condition, only Qwen3-14B uses the numbers (87.5–90.5%), and in the scale study Qwen3-8B and 32B do so as well (62% and 80% at $k = 8$; §5). A header stating that the list is “most recent first” is worse still (Qwen3-4B: 0–1%). A retraction marker attached to the wrong value (“ $t = 51$. Hmm, that’s wrong.”) helps but does not override position: when the retracted value comes last, Qwen3-4B still picks it in 60–80% of items, and the best model (Qwen3-14B) reaches 47.5% accuracy. In real reasoning traces (OpenR1-Math) with two counterfeit conclusions, the earlier one A and a later one B followed by “Wait, that’s wrong.”, a rational reader should choose A; with 10% of the derivation kept, readers still choose the retracted B at least as often (Qwen3-4B 56.5% vs. 19.4%, Qwen3-14B 45.5% vs. 27.5%, OLMo-2-13B-Instruct 41.5% vs. 40.0%). Reading its own unedited reasoning, by contrast, Qwen3-4B is accurate (93–100% on most BBH tasks): the positional default is usually right for text written in chronological order.

4.2 Deployed formats

Chat memory. We take the 78 knowledge-update questions of LongMemEval, each with two dated sessions (4.1k–9.3k tokens) in which the later session updates a fact from the earlier one, and present the sessions oldest or newest first (Figure 3a). Newest first costs Qwen3-4B 25.6 points (76.9→51.3), Qwen3-14B 24.4 and Phi-4-mini 52.6; stale answers rise to 41, 36 and 59%. All intervals exclude zero. Adding “the sessions are listed from the most recent to the oldest” recovers 0–2.6 points; removing the dates makes things worse, so the dates are used, but weakly.

Logs and agent memory. In synthetic e-mail threads, git logs and changelogs in which every entry is dated, newest-first presentation changes accuracy by $-12.1 [-15.6, -8.6]$ points for Qwen3-4B, -27.4 for OLMo-2-13B-Instruct and -31.4 for Phi-4-mini, and only -3.9 for Qwen3-14B in these short (0.8k-token) contexts. E-mail threads are hardest: Qwen3-4B drops from 88.3% to 58.0%, Phi-4-mini from 86.3% to 15.0%. In agent trajectories, inserting a retrieved, explicitly time-stamped stale memory right before the question lowers Qwen3-4B accuracy by 9.7 points ($k = 3$ updates: 88.7→69.3), while the same memory at the beginning of the context is harmless (-0.7): what matters is where the stale value sits.

4.3 Held-out benchmarks and retrieval order

ConvoMem-long. From ConvoMem’s changing-evidence questions [35] we take personas 50–99 (never used in any design decision) and the full 40-message conversations (median 2.4k tokens), date the two conversations in their source order, and present them oldest or newest first. Newest first changes Qwen3-4B’s accuracy by $\Delta = -17.7 [-25.0, -10.5]$ points (95.2→77.4) and Phi-4-mini’s by $-62.9 [-71.8, -54.8]$ (97.6→34.7).

MemConflict. MemConflict simulates users whose state changes over four years of monthly sessions and asks, after each update, how an attribute changed. We use dynamic-conflict questions in which the old value is *not* restated in the update session but appears in an earlier one, so that the answer requires relating two sessions by their dates. Of the 2,946 dynamic-conflict questions, 1,794 are of medium difficulty, 1,208 name exactly one changed attribute, 967 do not restate the old value in the update session, and 850 also mention it in an earlier session; we sample at most 8 per simulated user (240 questions). Each context contains the earlier session, the update session, and the sessions just before the update (2–6 sessions; median 18.5k tokens), each headed by its date. Qwen3-4B answers 75.4% correctly in chronological order, 36.2% newest first ($\Delta = -39.2 [-46.2, -31.7]$) and 56.2% in BM25 retrieval order ($\Delta = -19.2 [-25.4, -13.3]$); the share of answers that state the change

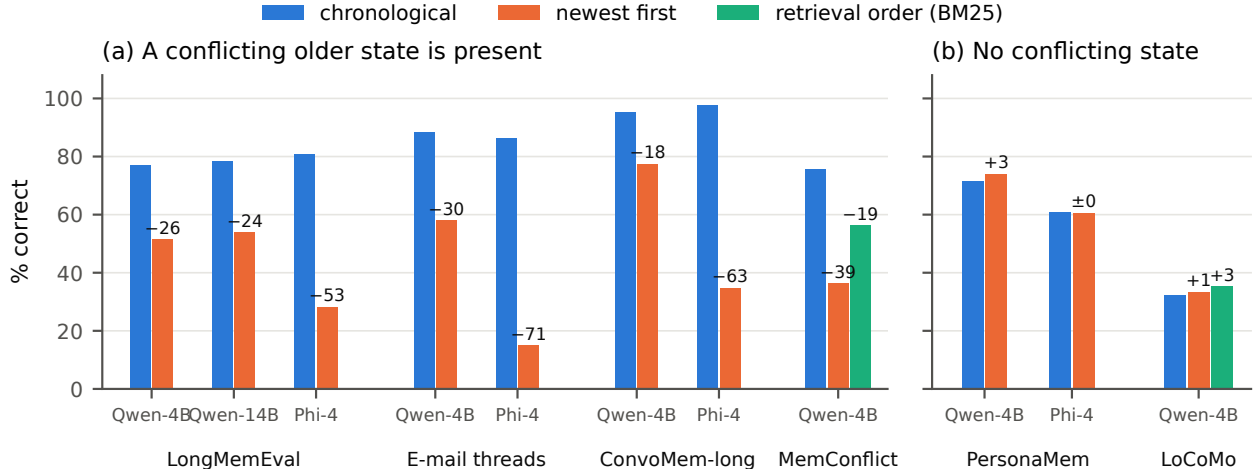


Figure 3: Order effects in deployed formats and third-party benchmarks (Qwen3-4B, Qwen3-14B and Phi-4-mini, HF). Numbers above bars are differences from the chronological bar. (a) When an older, conflicting state is present, newest-first presentation costs 18–71 points, and BM25 retrieval order costs 19 points on MemConflict. LongMemEval and e-mail numbers from §4.2; ConvoMem-long, MemConflict from §4.3. (b) PersonaMem and LoCoMo temporal questions, which do not require resolving such a conflict, show no order effect (§6).

in the wrong direction rises from 5.0% to 37.9% and 18.3%. For Phi-4-mini, $\Delta = -37.9 [-45.4, -30.8]$ newest first and $-14.6 [-20.8, -8.3]$ in retrieval order. Retrieval order places the update session first in 86 of 240 items; accuracy is 48.8% on these and 60.4% on the rest. Since the retrieved sessions are the same in both conditions, re-sorting retrieved memories by their timestamps before reading recovers the full 19 points at no cost.

The selection of items is not what creates the effect. The complementary questions, whose update session itself restates the old value (“moved from A to B”), form a pre-registered control (219 items after the same construction and sampling). The effect is smaller there but still large: newest first gives Qwen3-4B $\Delta = -24.2 [-31.1, -16.9]$, 15.0 [3.9, 26.6] points less negative than on the main set (user-clustered intervals), and retrieval order costs the same on both sets (difference +1.8 [−6.8, 9.8]). When the direction of a change is written inside one session, position matters less, but it still matters.

5 Scale and model family

We ran the same tests on Qwen3-4B, 8B, 14B and 32B in bf16 on one engine (Table 2, Figure 4). Before the remaining 32B results were in, we wrote down how a 32B result would change our claims (Appendix A): if the effect replicated on at least half of the sets on which it replicated at 4B, we would claim it for 1.7B–32B and report any attenuation; otherwise we would restrict the claim to models up to 14B.

The effect persists where dates are far from the facts. On MemConflict the newest-first cost is -40.4 , -37.1 and -35.8 points at 4B, 8B and 14B, and $-21.2 [-27.9, -14.6]$ at 32B, where 20.0% of answers still state the change in the wrong direction (3.3% in chronological order). The 32B effect is about half the 4B effect on the same items (+19.2 [11.2, 27.1]); chronological accuracy

Table 2: Order effect by model size (Qwen3, bf16, vLLM; same items, prompts and judge). Accuracy change (points) under newest-first (or retrieval) presentation relative to chronological, with 95% paired bootstrap intervals; CoT rows are percentages of items ($k = 8$, $n = 100$).

Test	Qwen3-4B	Qwen3-8B	Qwen3-14B	Qwen3-32B
MemConflict, newest first	-40.4 [-47.5, -33.3]	-37.1 [-43.8, -30.4]	-35.8 [-42.5, -29.2]	-21.2 [-27.9, -14.6]
MemConflict, retrieval order	-19.2 [-25.4, -13.3]	-17.9 [-23.8, -12.5]	-15.4 [-21.2, -9.6]	-10.0 [-15.4, -4.6]
chronological accuracy	77.1	78.3	84.6	80.0
LongMemEval, newest first (seen)	-24.4 [-35.9, -12.8]	-24.4 [-34.6, -14.1]	-23.1 [-33.3, -12.8]	-24.4 [-35.9, -14.1]
ConvoMem-long, newest first	-17.7 [-25.0, -10.5]	-12.1 [-17.7, -6.5]	-2.4 [-6.5, 1.6]	-6.5 [-11.3, -2.4]
Dated logs, newest first	-11.3 [-14.9, -7.9]	-13.6 [-17.1, -10.1]	-3.6 [-5.3, -1.8]	-2.9 [-4.7, -1.1]
Agent, stale memory before question	-9.3 [-12.7, -6.3]	-8.3 [-11.7, -5.3]	-0.7 [-2.0, 0.7]	-0.3 [-1.0, 0.0]
PersonaMem, newest first (no conflict)	0.0 [-3.4, 3.4]	-1.7 [-4.6, 1.3]	+3.8 [0.0, 7.6]	-1.7 [-5.5, 2.1]
CoT shuffled: picks last-presented (%)	98	99	97	97
CoT newest first + step numbers: correct (%)	4	62	63	80

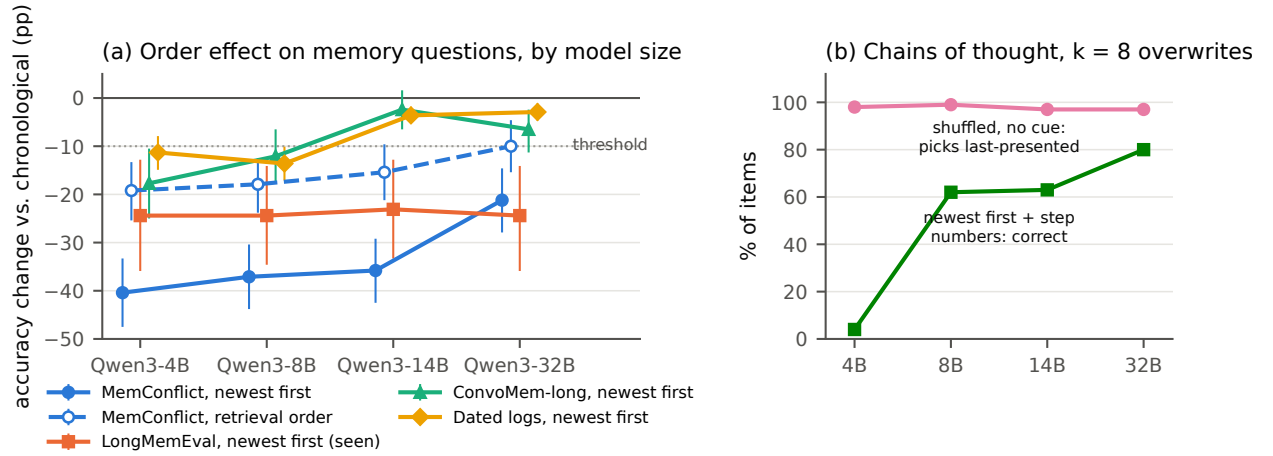


Figure 4: Scale removes the effect only where dates are local (Qwen3, bf16, vLLM). (a) Newest-first (and retrieval) minus chronological accuracy with 95% paired bootstrap intervals; the dotted line is our replication threshold. Where every record carries its own date (logs) or the context is short (ConvoMem-long, 2.4k tokens), the effect fades by 14B; where dates sit in session headers thousands of tokens before the facts (MemConflict, LongMemEval), it persists at 32B. (b) Chains of thought: without a time cue, all sizes read the last-presented value; with a step number on every line, larger models increasingly use it.

is similar (80.0 vs. 77.1), so the difference is not a ceiling effect, and the judge is as reliable on 32B’s answers as on 4B’s (§3). Retrieval order still gives $\Delta = -10.0 [-15.4, -4.6]$ at 32B, exactly at our threshold, and accuracy is again lower when the update session is retrieved first (62.8% vs. 74.0%). On LongMemEval the cost does not change with size (-23.1 to -24.4), and at 32B a “most recent first” header still leaves $\Delta = -17.9 [-28.2, -9.0]$. The 32B effect does not depend on length within MemConflict (terciles with median 14k, 18k and 21k tokens: $-21.2, -22.5, -20.0$).

Where dates are local, scale removes it. On dated logs, agent memories and the 2.4k-token ConvoMem-long conversations, the effect falls below 7 points at 14B and 32B (Table 2); on ConvoMem-long, 32B is at ceiling in chronological order (100%). In chains of thought, larger models use step numbers on newest-first lists (Qwen3-32B: 80% at $k = 8$, 4B: 4%) and “most recent first” headers (32B: 59%), but without any cue every size reads the last-presented value (97–99%). We had predicted this pattern before seeing the remaining 32B results (effect below 5 points on dated logs, at least 10 points on long sessions with dates in headers); it held for logs, MemConflict and

LongMemEval and failed for ConvoMem-long, whose conversations are short and on which 32B is at ceiling.

Verdict by the rule written in advance. Of the sets on which the effect replicated at 4B, it replicates at 32B on MemConflict (both orders), LongMemEval and shuffled CoT, and not on ConvoMem-long or dated logs; by our rule, we claim the effect for 1.7B–32B and report that it weakens with scale in short or locally dated contexts. Qwen3-14B in 4-bit on HF gives the same picture as in bf16 on vLLM (MemConflict -33.8 vs. -35.8).

A second family. Gemma 3 12B shows the effect on MemConflict more strongly than Qwen3 of similar size [pending: judge-scored results and Gemma 3 27B]. With a string rule that we calibrated on Qwen (it understates the judge-scored effect by about 5 points), newest first gives Gemma $\Delta = -52.5 [-59.2, -45.8]$ on MemConflict (Qwen3-14B: -24.2 by the same rule), with 57.5% of its answers naming the old value as the new one, and retrieval order gives $-20.0 [-25.8, -14.2]$. Unlike Qwen at 14B and 32B, Gemma is also affected by newest-first changelogs ($99.3\% \rightarrow 65.7\%$), whereas PersonaMem again shows no effect (-0.4). Gemma’s CoT reading is less positional (the last-presented value is chosen in 66% of shuffled items at $k = 8$).

6 Where order does not matter

PersonaMem and LoCoMo. On the 238 PersonaMem questions about how a preference evolved and why it changed (16k–30k tokens), presenting sessions newest first has no effect (Qwen3-4B $+2.5 [-0.8, 5.9]$; Phi-4-mini $-0.4 [-3.8, 2.9]$); the same holds for the five other question types (largest drop -5.9 , $n = 17$), for every Qwen3 size up to 32B (-1.7 to $+3.8$; Table 2) and for Gemma 3 12B, and for the 321 LoCoMo temporal questions (Qwen3-4B $+1.2 [-3.7, 5.9]$ newest first, $+3.1 [-1.6, 7.8]$ in retrieval order; Qwen3-32B $-2.8 [-7.2, 1.6]$ and $-2.5 [-7.5, 2.5]$). Diagnostics explain why (Appendix B). PersonaMem’s update questions can be answered without the history: matching words between the user’s final message and the options alone selects the correct option for 80.8% of the “why did the preference change” questions (Qwen3-4B with the full history: 83.8%), and Phi-4-mini answers “how did the preference evolve” questions equally well with and without the history (51.8% both). One cell does not fit cleanly: by the criterion we wrote in advance (accuracy without the history at least 10 points below accuracy with it), Qwen3-4B’s “why did it change” questions do depend on the history (-12.1 points without it, but only when a persona description is shown; with the final message and options alone it scores 80.8% against 83.8%), and newest-first order still does not hurt; we report this cell as unresolved rather than explained. The remaining question types do depend on the history, but none requires choosing between an older and a newer state; neither do LoCoMo’s temporal questions, which ask when an event happened (we have not validated the judge on LoCoMo’s free-form date answers as we did on the other sets). These null results delimit the phenomenon rather than contradict it: order matters when a conflicting older state is present and time is expressed only by position or by dates far from the facts.

A confound. On MemoryAgentBench’s conflict-resolution facts [19], where the newer fact is counterfactual and the older one is world knowledge, Qwen3-4B picks the older fact in 83% of chronological items and Qwen3-32B in 82%: parametric knowledge dominates and the set cannot test order, in its 6k- or its 37k-token version (it was also used in our pre-registered training evaluation, so we treat it as seen). The BEAM benchmark [43] could not be used either: in 168 of its 180

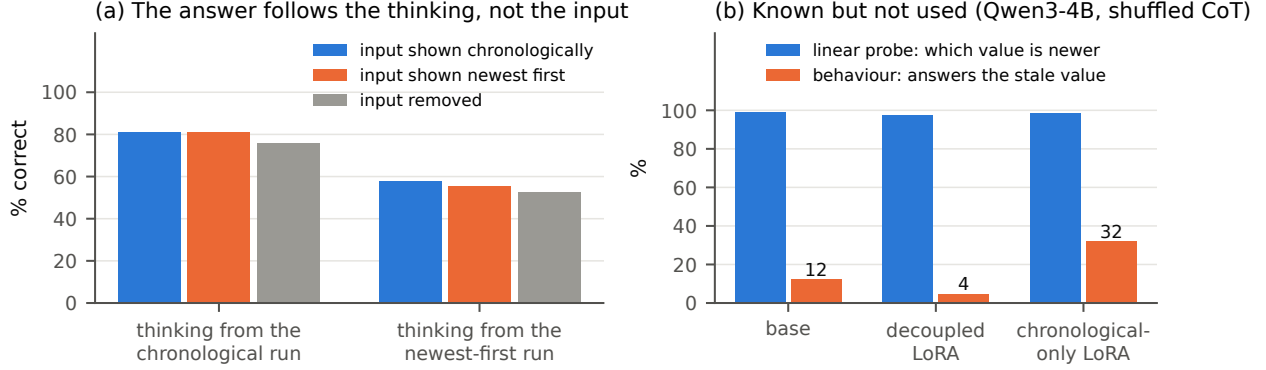


Figure 5: Mechanism. (a) Qwen3-4B in thinking mode on LongMemEval ($n = 78$): accuracy depends on which run the thinking comes from, not on the order of the input it is paired with, or on whether the input is shown at all. (b) On shuffled, step-numbered CoT, a linear probe at the answer position decodes which value is newer in $\approx 98\%$ of items for the base model and both LoRA variants; behavior differs widely. Training changes use, not knowledge.

knowledge-update questions the old and the new information share one dated session, so no explicit time separates them.

7 Why: two failure points

7.1 Misleading order is copied into the model’s own reasoning

A natural defense is to let the model reason first. It does not help, because the reasoning inherits the presentation order.

Prefilled timelines. We write a correct, dated timeline of the two pieces of evidence into the model’s response and let it only produce the final answer (Qwen3-4B, Qwen3-14B, $n = 70$). When the timeline lists the evidence newest to oldest and the input is newest first, both models still answer the stale value in about 40% of items (40.0% and 38.6%; Phi-4-mini: 57.1%), although every date in the list is correct; with an oldest-to-newest list the rate falls to 27.1% and 10.0%. Left to their own devices, models mostly list evidence in presentation order.

Swapping thoughts. In thinking mode on LongMemEval, Qwen3-4B answers 80.8% correctly from chronological input and 55.1% from newest-first input (Figure 5a). Its thinking restates the evidence in presentation order: among traces that mention both session dates, the newer date comes first in 82.1% of newest-first runs and 3.6% of chronological runs. Swapping the thinking between runs swaps the accuracy: newest-first input with the thinking from the chronological run gives 80.8%; chronological input with the thinking from the newest-first run gives 57.7%; removing the input entirely changes little (75.6% and 52.6%). The answer follows the thinking, and the thinking has absorbed the misleading order. In e-mail threads, thinking fixes the order in short contexts (10 messages: 93.3% correct newest first) but not in longer ones (200 messages: 53.3%), where the last value mentioned in the thinking is the newest one in only 51.7% of items.

7.2 Short contexts: known but not used

We train linear probes on the residual stream at the last token before the answer to predict whether the newer value was presented first (layers selected on training items only; 60/40 split by question). On shuffled, step-numbered CoT the probe reaches 98.8% (layer 21), and it decodes the correct order on all 13 items that Qwen3-4B answers with the stale value. On ConvoMem short dialogues it reaches 97.9%; Qwen3-14B answers MemoryAgentBench fact pairs stale in 50% of items while its probe reaches 94%. The representation is format-specific: probes transfer across formats at near chance (49–58%). In short contexts, the model knows which value is newer and fails to select it.

7.3 Long sessions: dates are not bound to facts

In long LongMemEval sessions, the same probe is weak (best layer 68.6% for Qwen3-4B and 72% for Qwen3-14B in five-fold cross-validation), suggesting that “which update is newer” is not available at the answer position. Models that we trained to write a dated list before answering (Appendix D, first version) name both session dates in only 19.2% of newest-first items; when they do, they answer correctly in 86.7% of items, otherwise in 46.0%. The dates sit in session headers thousands of tokens before the facts they govern, and the model does not attach them. The information is nevertheless readable: forcing attention onto the newer statement at the answer position, an oracle intervention that requires knowing where the answer is and that we use only as a diagnostic [following 16], raises newest-first LongMemEval accuracy from 52.9% to 72.9% (e-mail threads: 60.7% $\rightarrow$ 100%). The scale study fits this account: larger models stop falling back on position when each record carries its own date (logs, step-numbered CoT), but not when the date sits in a header far from the fact (§5). This is a post-hoc reading of a correlational pattern, not a test.

7.4 A rule, or recency?

Every test above asks for the *current* value, so a reader that maps later text to later time and a reader that simply favors whatever is closest to the answer make the same prediction. Questions about the *earliest* value separate them: on newest-first input, a semantic “written later means newer” rule picks the last-presented record for the current value and the first-presented one for the earliest value, while mechanical recency picks the last-presented record for both. In a pre-registered test (dated calendar and audit tables with 2–4 records, 150 items per cell, two formats never used elsewhere), no model meets the semantic criterion (current: last minus first ≥ 20 points; earliest: ≤ -20): Phi-4-mini favors the last-presented record for both questions (+36.7 and +18.0, just short of the mechanical criterion), Qwen3-8B does so for the earliest value only (+2.7 and +28.0), Qwen3-4B and Gemma 3 12B are mixed, and Qwen3-14B and 32B mostly answer by date in these short tables (current -53.3 and -54.0 , earliest $+53.3$ and $+56.7$), although newest-first order still costs 32B 20.0 points on the current value. We therefore do not claim that models interpret later text as later time; we claim that, when they do not use explicit time, they fall back on position.

8 Fixing the failure

8.1 Prompting does not help

Stating the order (“listed from the most recent to the oldest”) recovers at most 2.6 points on LongMemEval in Qwen3-4B, and 6.4 in Qwen3-32B. Asking the model to first write a timeline sorted by date leaves a 21.8 [11.5, 32.1]-point stale-rate gap between newest-first and chronological input. Thinking mode reaches 52.6% newest first with a sampled answer (55.1% with a greedy answer from

the same kind of thinking), little better than non-thinking (51.3%); answering only from the model’s own notes gives 52.6%. Because the failure lives in the reasoning itself, instructions to reason do not remove it. [pending: base model with thinking on, on the held-out sets (ConvoMem-long, MemConflict, PersonaMem)]

8.2 Decoupled training changes use, not knowledge

Data. We generate 5,000 synthetic training samples (9.4M tokens) in formats that appear in no test: sticky notes, ticket updates, CSV change logs, key–value configuration snapshots, ledgers, sensor reports and meeting minutes. Each sample updates one attribute $k = 1\text{--}6$ times among distractor records and filler prose (0.3k–9.5k tokens). Records of the same attribute use templates drawn from one pool, contain no cue words (“now”, “updated”, “latest”), and differ only in their dates, written in mixed formats. Questions ask for the current value (60%), the earliest value (20%) or the value as of a date (20%), so that “always pick the newest date” is not a shortcut. In 20% of samples there are no dates; the answer is then the last record, or the first if the text states that it is ordered newest first. In the **decoupled** condition records are presented chronologically, reversed or shuffled (one third each), so position carries no information about time. The **control** contains the identical samples, always chronological; we verified that the two training sets differ only in record order.

Training and evaluation. We fine-tune Qwen3-4B with LoRA [18] (rank 16, $\alpha = 32$, learning rate 10^{-4} , one epoch, final checkpoint), computing the loss on answer tokens only. Hyperparameters were fixed on a development set in two further formats (calendar entries, audit tables) before any test was run; the evaluation scripts were frozen and hashed beforehand.

Results. Decoupled training transfers to formats it never saw (Table 3, Figure 6a). On newest-first logs it reaches 100% (control 89.3%, base 70.3%). On newest-first chains of thought with step numbers it reaches 74.7% (control 2.0%, base 6.0%), although training contained dates and never step numbers. It repairs reading of the model’s own lists: with a prefilled newest-to-oldest timeline and newest-first input it answers 77.1% correctly (control 51.4%; +25.7 [14.3, 37.1]); when the model writes the timeline itself the gain is +19.2 [9.0, 29.5], and Phi-4-mini shows the same pattern (+27.1 [14.3, 40.0]). The probe analysis (Figure 5b) shows what changed: all three models represent which value is newer ($\approx 98\%$), but the stale rate is 12.2% for the base model, 4.5% after decoupled training and 31.8% after chronological-only training, whose probe still decodes the correct order on 47 of the 48 items it answers stale. Decoupled training improves the use of a representation the model already has; chronological-only training suppresses it. Training only on decoupled chains of thought transfers in the other direction, to dated logs (+28.7 [24.7, 32.9] over its control), suggesting a single skill. The same pre-registered pair at Qwen3-1.7B transfers as well (newest-first logs: base 29.3%, control 1.4%, decoupled 78.3%; step-numbered newest-first CoT: 1.7%, 0.0%, 27.3%) [pending: 1.7B LongMemEval endpoint].

Limits: not a usable fix. The pre-registered primary endpoint was not met: on newest-first Long-MemEval sessions the decoupled model scores 48.7% (control 44.9%, base 51.3%). On TempReason [40], which requires matching dates to intervals, decoupled training scores $-7.0 [-11.7, -2.7]$ points relative to the control (1.7B: -10.7), possibly an over-preference for the latest record. Because every training target was a bare value, both LoRAs also stopped following answer-format instructions: on our memory test, which asks for one sentence, 55.1% (decoupled) and 46.4% (control) of the answers have at most three words, against 0.9% for the base model. At 4B, GSM8K [10] is unaffected

Table 3: Decoupled answer-only training (pre-registered; Qwen3-4B, seed 0). Accuracy (%) under non-chronological presentation; Δ = decoupled – chronological-only control with 95% paired bootstrap interval. No test format appears in training. MemoryAgentBench and LongMemEval are *seen* (§3). [†]Exploratory follow-up (E2), not part of the pre-registered test list.

Test	Condition	Base	Control	Decoupled	Δ
Logs (git, CHANGELOG, e-mail)	newest first	70.3	89.3	100.0	+10.7 [8.7, 12.7]
CoT, step numbers	newest first	6.0	2.0	74.7	+72.7 [67.7, 77.7]
CoT, step numbers	shuffled	34.7	27.3	45.7	+18.3 [14.0, 22.7]
Prefilled list (new→old) [†]	newest first	57.1	51.4	77.1	+25.7 [14.3, 37.1]
MemoryAgentBench, single hop	shuffled	42.0	57.0	70.0	+13 [4, 23]
MemoryAgentBench, multi hop	newest first	5.0	4.0	19.0	+15 [8, 23]
LongMemEval sessions	newest first	51.3	44.9	48.7	+3.8 [−3.8, 11.5]
Test of Time	shuffled	28.0	29.2	32.4	+3.1 [−2.9, 9.2]
TempReason	newest first	70.3	78.3	71.3	−7.0 [−11.7, −2.7]

(90.4% base, 92.4% both LoRAs; the models still reason step by step), but at 1.7B both LoRAs stop reasoning (median answer length 57 words before, 2 after) and GSM8K falls from 78.4% to 19.6% (decoupled) and 26.0% (control), as does BABILong’s two-supporting-facts task [23] (31.3%→0.7% and 2.7%); our pre-registered limit for general ability (≤ 2 points) is badly missed. We therefore use this training only as evidence about the mechanism; the selection failure does not reach long sessions, where dates are not bound to the facts.

8.3 A dated list inside the model’s thinking

The long-session failure needs the reader to attach each fact to its date before comparing. Writing the relevant records with their dates is the step that works when the model does it (above: 86.7% correct when both dates are written, 46.0% otherwise), and a diagnostic showed that a list is needed only when dates are far from the facts (on LongMemEval, which is seen, a variant trained to answer directly scored 16.7 [−26.9, −6.4] points below one that wrote the list first; on ConvoMem-long they did not differ; Appendix D). Our fix therefore teaches the model to write the list in its thinking, so that the visible answer can follow any requested format.

Data. We generate 2,500 long synthetic documents organized in blocks, each opened by a single date line and followed by 150–1500 tokens of unrelated prose in which the statements are buried at random depths (genres: diary, meeting minutes, lab notebook, status report, field log; no chat, e-mail or log formats; median 2.8k tokens). Block order is decoupled from time (chronological, reversed or shuffled). Questions ask for the current value, the earliest value, the value as of a date, the most recent change, the whole history of a change, and the reason for a change; others offer multiple-choice options (values or full sentences), or concern attributes recorded once or never. Answer instructions vary (value only, one sentence, with the date, letter only), with different wordings in training and in the development set. In half of the samples, the target thinking lists only the queried attribute’s records, oldest to newest with their dates, followed by one sentence of inference; for reason questions it quotes the relevant record without a list. The other half has no thinking. The set also contains 1,000 samples with local dates from an earlier run and 474 general instruction responses that the base model wrote itself for prompts from the Tulu 3 mixture [25], excluding instruction-following, math word-problem and benchmark-related sources (11.9% of 3,974 samples), following the practice of mixing general data into long-context training [2]. We train Qwen3-4B with the same LoRA

Table 4: Dated list in thinking (Qwen3-4B, seed 0). Accuracy (%), chronological / newest first (/ retrieval). Base without thinking on HF; E18 on HF (desktop), E18.1 on HF (cloud); thinking-mode decoding as in §3. *Aggregate failure types on this set informed the design of E18.1 (Appendix D). LongMemEval is seen.

Test	Base	E18, thinking off	E18, thinking on	E18.1, thinking on
ConvoMem-long (held out)	95.2 / 77.4	97.6 / 87.9	97.6 / 96.8	96.8 / 96.8
MemConflict (held out*)	75.4 / 36.2 / 56.2	71.7 / 68.8 / 70.0	72.9 / 71.7 / 70.0	80.4 / 77.5 / 77.9
PersonaMem (held out*; no conflict)	71.4 / 73.9	72.7 / 72.3	63.9 / 56.7	71.4 / 71.0
LongMemEval, newest first (seen)	51.3	51.3	61.5	[pending: running]

settings as above, with the loss on the target thinking and answer (one epoch, 486 steps, final checkpoint).

Results. With thinking on, the order effect disappears on the held-out long-dialogue sets (Table 4, Figure 6b). On ConvoMem-long, the only held-out set whose failure types we never analyzed when designing the fix, newest-first accuracy rises from 77.4% to 96.8% and equals chronological accuracy ($\Delta = 0.0 [-4.0, 4.0]$). On MemConflict it rises from 36.2% to 77.5% newest first and from 56.2% to 77.9% in retrieval order, while chronological accuracy does not fall (75.4 $\rightarrow$ 80.4%); 8.3% of newest-first answers are stale (base: 37.9%) and 0.1% of the thinking runs out of budget. PersonaMem accuracy stays within 3 points of the base model (71.4/71.0% vs. 71.4/73.9%; threshold: 5 points), the main threshold of this version. These results are from the fifth version that we evaluated on held-out data (Appendix D); MemConflict and PersonaMem informed its design through aggregate failure types, so the cleanest evidence is ConvoMem-long.

What the previous version shows. The first thinking-mode version (E18) passed every general-ability threshold we set in advance with thinking off (MMLU [17] +0.5, ARC-Challenge [9] +1.5, HellaSwag [49] -0.2, IFEval [51] +1.8, LongBench [3] four-task average +2.3, GSM8K -1.6 [-5.6, 2.4]; lm-evaluation-harness [5]), but with thinking on it lost 7.5 and 17.2 points on PersonaMem: in 98% of its thoughts it applied the trained “list the records, take the newest” template even to questions about reasons and about the evolution of a preference, and 10.9% of its thoughts did not finish. Adding reason questions, sentence options and self-distilled general data (E18.1) removed this harm. E18 also shows that most of the gain on MemConflict does not come from thinking mode: with thinking off, it already answers 68.8% correctly newest first (base without thinking: 36.2%). On ConvoMem-long, training alone recovers part of the gap (87.9%) and thinking the rest (96.8%); on LongMemEval (seen), only thinking mode helps (61.5 vs. 51.3%).

Open items. By our pre-registered rule, E18.1 did not pass the format test on synthetic development data (each answer instruction followed in $\geq 90\%$ of items): with thinking off, value-only answers reach 88.2%, letter-only 89.5% and dated answers 76.2% (81.0% with thinking). All 22 failing answers were correctly formatted; the content was wrong, half of them on as-of-date questions, on which E18 scores the same (63.3%) and the base model 30.0%. We report the threshold as not met. Still to come: general ability for E18.1 with thinking on and off [pending: running], seeds 1 and 2 [pending: running], the base model with thinking on as a control on the held-out sets [pending: running], a chronological-only control for this recipe [pending: running], LongMemEval and e-mail threads [pending: running], and the same recipe on Qwen3-8B [pending: running].

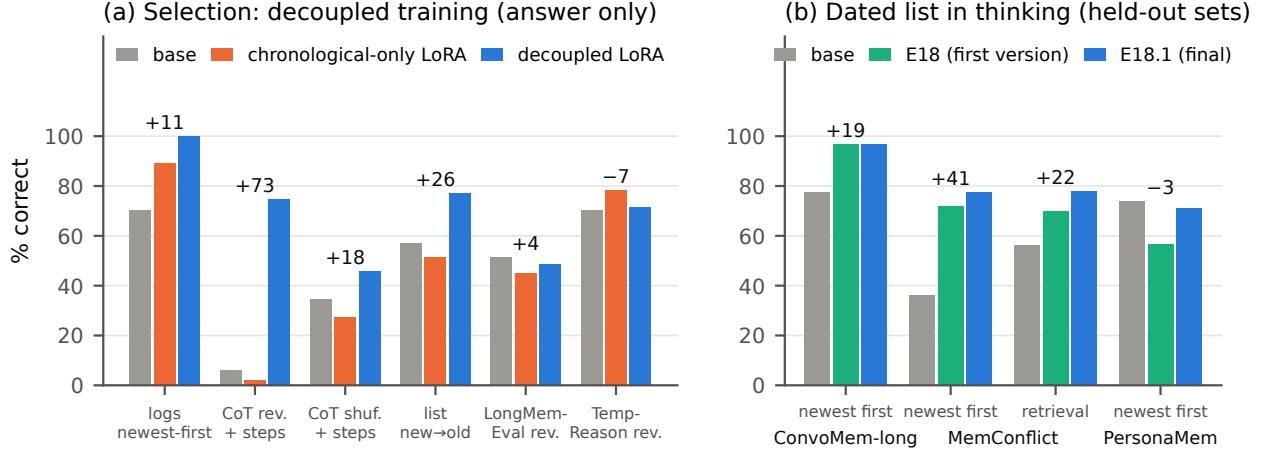


Figure 6: Training. (a) Decoupled versus chronological-only answer-only LoRA (Qwen3-4B, pre-registered); numbers are decoupled minus control. Transfer to unseen formats, failure on LongMemEval sessions and a cost on TempReason; this training also damages answer formats (§8.2). (b) Dated list in thinking, thinking on, against the base model without thinking; numbers are E18.1 minus base. The first version (E18) repaired the order effect but lost 7–17 points on PersonaMem, where no conflicting state is present; E18.1 stays within 3 points.

9 Discussion and limitations

What the model does. Our results suggest a reading strategy that is reasonable for the data models are trained on: when several values compete and explicit time is not used, take the one written last, because text is almost always written in the order things happen. Explicit time is used when it is local and salient, increasingly so with scale (step numbers, dated log entries), but in long contexts it is not bound to the facts and position decides, at 4B as at 32B. The strategy fails exactly when presentation order and time disagree and a conflicting older state is present, which is common in deployed systems.

Practical implications. Memory and retrieval pipelines should present retrieved evidence in chronological order whenever timestamps are available; on MemConflict this recovers 10–19 points at every size we tested, at no cost. Clients that show newest-first threads or logs to a model should reverse them, at least for models below 14B. Statements about the order, instructions to reason, and thinking mode are not substitutes.

Limitations.

- **Scale.** Our largest model has 32B parameters, and the second family is preliminary (one size, string scoring). The effect weakens with scale on some sets; we do not extrapolate beyond 32B, and we have not tested API models.
- **Engines.** The scale study runs on vLLM and the 4B, 14B and Phi-4-mini application experiments on HF; item-level agreement between the engines is 94–97%. We compare models within an engine and mark cross-engine comparisons.
- **Held-out reuse.** We evaluated five versions of the long-session fix (eight trained models, counting controls) on the held-out sets during development, and MemConflict and PersonaMem informed later versions through aggregate failure types. We never analyzed ConvoMem-long’s failure types, but it is small (124 items) and near ceiling for larger models.

- **The fix is not finished.** E18.1 is one seed of one 4B model; its general ability, the base model with thinking on, and a chronological-only control are still running (§8.3). It missed our format threshold on as-of-date questions. Writing a list costs tokens, and at 25k tokens the first thinking version produced lists that did not finish.
- **Seeds and exploratory status.** The answer-only training is pre-registered but reported for one seed per model; at 1.7B it damages general ability. The mechanism experiments are exploratory, with thresholds written before running.
- **Judges and parsing.** Free-text answers are graded by Qwen3-14B (about 91% agreement with string matching where both apply). For PersonaMem we corrected a parser after seeing that it failed on answers ending in “The best answer is (d).”; the rule uses only the format, and no prediction of the base models changed.
- **Constructed orders.** Newest-first presentation is constructed by us, although it mirrors common interfaces; retrieval order is not, and it costs 10–19 points.

10 Conclusion

When presentation order and explicit time disagree, language models follow the order. The default is cheap and usually right, which is why it survives explicit dates, step numbers, ordering statements, retractions and plain thinking; scale teaches models to use dates that sit next to the facts, but not dates in headers thousands of tokens away. The failure has two parts, selecting the newer value and binding facts to their dates, and the second can be trained into a model, without seeing the formats where it is tested, by teaching it to write down the dated records before it answers.

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A Pre-registered thresholds and outcomes

Thresholds were written in `EXPLORE_PLAN.md`, `PREREG_TRAINING.md` or `CLOUD_NOTEBOOK.md` before each run. Experiments are Qwen3-4B, seed 0, unless noted. “Met” and “not met” refer to the written threshold.

Experiment	Threshold (written before running)	Outcome
Selection LoRA, primary end-point	LongMemEval newest first: decoupled $\geq$ control +10, CI > 0	Not met (+3.8)
Selection LoRA, transfer	two further test families, CI > 0	Met (logs, CoT, MAB)
Selection LoRA, side effects	chronological conditions and general ability drop ≤ 2	Met at 4B; TempReason -7.0 vs. control; not met at 1.7B (GSM8K -58.8)
E1 prefilled timelines	newest-first input raises stale rate ≥ 15 for both models	Met (+28.6 each)
E1c thinking (first reading)	thinking conclusions ≥ 15 more stale	Not met; corrected by swap experiment (grader read only the last 800 characters)
E1c swap	answer follows thinking	Met
E1d sorted timeline instruction	stale rate ≥ 15 below the plain timeline prompt and gap to chronological ≤ 5	Not met (gap +21.8)
E2 list reading, Qwen3-4B	decoupled $\geq$ control +10	Met (+25.7; own timeline +19.2)
E2 list reading, Phi-4-mini	same	Met (+27.1)
E3 zeroing recency heads	drop $\geq 2\times$ random and ≥ 10	Not met; confounded (reading also breaks)
E4 primacy at length	first-value rate +20 up to 19k tokens	Not met (0%)
E5 CoT-only LoRA	≥ 10 on two application tests	Met; LongMemEval reversed (-20.5)
E7 probes, short contexts	$\geq 80\%$ on stale-answered items	Met (100%, $n = 13$)
E7 probes, long sessions	$\geq 80\%$ or $\leq 60\%$	Inconclusive (69%)
E10 attention routing (oracle)	+15 on a long-dialogue set	Met (LongMemEval +20)
E11 answer from notes only	$\geq$ thinking +15	Not met
New long-dialogue sets	newest first $\leq$ chron. -10 , CI < 0	ConvoMem-long met; PersonaMem not
External sets	same	MemConflict met (4B, Phi); LoCoMo not (predicted); MAB-CR untestable
E15 earliest-value questions	semantic: current $\geq +20$ and earliest ≤ -20 ; mechanical: both $\geq +20$	Mixed for all six models (Qwen3 4B–32B, Phi-4-mini, Gemma 3 12B)
E16 restated-value control	restated $-$ main $\geq +15$, CI > 0	Met, at the threshold (+15.0)
Scale: 32B replication	same rule per set	Met: MemConflict (both orders; retrieval at -10.0), LongMemEval, CoT; not: ConvoMem-long (ceiling), logs
Scale: claim at 32B	replicate on $\geq$ half of the 4B-replicated sets	Met $\rightarrow$ claim for 1.7B–32B, weaker with scale
Scale: prediction	32B logs < 5; long header-dated sessions ≥ 10	Met for logs, MemConflict, LongMemEval; not for ConvoMem-long
Fix diagnostic: list needed?	answer-only variant ≥ 10 below list on a long set	Met (LongMemEval -16.7 ; ConvoMem-long -0.8)
E18	format ≥ 90 ; ConvoMem-long; PersonaMem $\geq$ base -5 ; MemConflict; general ability	Met except PersonaMem with thinking on ($-7.5/-17.2$) and one format cell with thinking off (89.7)
E18.1	PersonaMem $\geq$ base -5 ; keep ConvoMem-long and MemConflict; format ≥ 90	Applied tests met; format not met (as-of-date content); general ability [pending: running]

B PersonaMem diagnostics

Table 5 reports accuracy without the history (*bare*: final user message and options; *no history*: persona, message and options) and with the full history in both orders. By the threshold written in advance, a question type “does not depend on the history” if chronological accuracy exceeds no-history accuracy by less than 10 points or the interval includes zero. “How did the preference evolve” questions do not depend on the history for either model. For “why did it change” questions, Qwen3-4B’s no-history condition is 12.1 [4.0, 20.2] points below the full history, which formally counts as dependence; the bare condition (80.8%), however, is within 3 points of the full history, and the persona text lowers accuracy. We note that this reading was made after seeing the results.

Table 5: PersonaMem 32k accuracy (%) by question type. Bare / no history / chronological / newest first.

Question type	n	Qwen3-4B	Phi-4-mini
Track preference evolution	139	56.8 / 58.3 / 62.6 / 67.6	51.8 / 51.8 / 51.8 / 51.1
Reasons behind updates	99	80.8 / 71.7 / 83.8 / 82.8	68.7 / 70.7 / 73.7 / 73.7
Recall user-shared facts	129	40.3 / 48.1 / 59.7 / 56.6	31.0 / 41.1 / 46.5 / 48.1
Recall facts mentioned	17	11.8 / 17.6 / 58.8 / 52.9	41.2 / 41.2 / 70.6 / 70.6
Generalize to new scenarios	57	21.1 / 38.6 / 63.2 / 57.9	17.5 / 28.1 / 43.9 / 38.6
Preference-aligned recommendations	55	40.0 / 23.6 / 58.2 / 61.8	34.5 / 34.5 / 47.3 / 47.3
Suggest new ideas	93	12.9 / 10.8 / 10.8 / 9.7	15.1 / 12.9 / 22.6 / 17.2

C Test-set construction

ConvoMem-long. ConvoMem changing-evidence items with two evidence conversations, personas 50–99, four items per persona (124 items). Both full conversations are shown, headed “Conversation (date: ...)” with dates drawn so that the earlier conversation in the source gets the earlier date; a “Current date” line follows, and the model answers in one sentence.

MemConflict. From the released benchmark file (30 simulated users), dynamic-conflict questions of medium difficulty whose reference answer names both the old and the new value of exactly one changed attribute. We keep a question only if the old value does not occur in the update session and does occur in an earlier session (structural filter, no model outputs used), take the latest such session, the update session and the sessions immediately before the update up to about 20k source tokens, and sample at most 8 questions per user. A few malformed messages in the source (28 of 142k) are rendered by their available text. The restated-value control uses the same construction on the questions whose update session mentions the old value.

PersonaMem. PersonaMem-v1, 32k contexts. The context up to the question is split into sessions at the persona system message; sessions receive increasing dates in source order and are shown oldest or newest first; four-way multiple choice.

LoCoMo. All 321 temporal questions (category 2) over the full conversations, each session headed by its original date and time.

Retrieval order. BM25 ($k_1 = 1.5$, $b = 0.75$) between the question and each session (or fact), computed within each item; ties keep source order.

Long prompts on vLLM. Four Test of Time prompts exceed Qwen3’s 40,960 positions and are skipped on vLLM; comparisons across engines use the common items.

D Versions of the fix evaluated on held-out data

We report every version of the long-session fix that we evaluated on a held-out set, in order, with what changed and why. Only E18.1 is reported as our fix; the others are development history.

- **E12** (with a chronological-only control): answer after a dated list, trained on records with local dates. ConvoMem (short, personas 50–99): newest first +2.4 over its control and +9.7 over the base model, below the threshold of +10 for both. The control learned to restate evidence in presentation order.
- **E13** (decoupled and chronological-only arms) and **E13b** (answer only): long block-dated documents, list then a bare value. ConvoMem-long newest first 95.2 and 96.0 (base 77.4); on MemConflict the bare answers did not state the change (chronological accuracy fell 9.2 and 5.4 points), and PersonaMem fell 11–18 points. Dropped because it did not follow answer formats. E13b served only as the diagnostic in §8.3.
- **E17**: same documents; varied question types and answer formats. ConvoMem-long 96.8/96.8; MemConflict 72.5/72.5/73.3 (first version designed with MemConflict’s aggregate failure types); PersonaMem 60.1/57.6, because it always wrote the list, even when asked for a letter.
- **E18**: list moved into the thinking block; thinking on and off; multiple-choice, single-record and absent-attribute questions (designed with PersonaMem’s aggregate failure types). See §8.3.
- **E18.1**: adds reason questions, sentence options and 11.9% self-distilled general data. See §8.3.

E Training data details

[To be expanded: generators `gen_train_data.py` (decoupled answer-only training), `gen_bind_data_v4.py` and `gen_selfdistill.py` (E18.1); data hashes (E18.1 train 0B46429F..., val 12CF83BB..., dev CDFADCCF..., LF); development-set curves; the full pre-registration with its change log.]